ISOSORBIDE MONONITRATE

Never with PDE-5 inhibitors

Morning dose

CHEMICAL NAME 1,4:3,6-dianhydro-D-glucitol 5-nitrate
CLASS Organic nitrate (long-acting antianginal)
TRADE NAMES Imdur · Monosorb · Elantan · Ismo

THERAPEUTIC USES / INDICATIONS

- Prophylaxis of stable angina.
- Adjunct in chronic heart failure.

ADMINISTRATION

- **PO:** IR twice daily (8 am and 2 pm); MR once daily in the morning — swallow whole.
- Not for acute attacks.

DOSAGES

- **IR:** 10–20 mg twice daily (second dose 8 h after first); up to 120 mg/day in divided doses (asymmetric, nitrate-free interval).
- **MR:** 30–60 mg once daily in the morning; max 120 mg/day.
- Start low (MR 30 mg) to reduce headache.

PHARMACOKINETICS

- **Onset:** 30–60 min. **Duration:** IR 6–8 h; MR 12–24 h.
- No first-pass metabolism (bioavailability ~100%).
- Excreted in urine. **t½:** 4–5 h.

PHARMACODYNAMICS

- NO donor → ↑ cGMP → venodilation (↓ preload), arterial and coronary dilation.
- ↓ myocardial oxygen demand.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache (common early).
- Dizziness, flushing.
- Nausea.

ADVERSE EFFECTS (SERIOUS)

- Hypotension, syncope.
- Tolerance if no nitrate-free interval.
- Reflex tachycardia.

CONTRAINDICATIONS & CAUTIONS

- PDE-5 inhibitor use.
- Severe hypotension, hypovolaemia.
- HOCM, aortic stenosis, constrictive pericarditis, tamponade.

KEY INTERACTIONS

- **Sildenafil, tadalafil, vardenafil:** contraindicated.
- **Antihypertensives, alcohol:** ↑ hypotension.

NURSING MANAGEMENT

- Check BP; give MR in morning.
- Make sure patient also has SL GTN for acute attacks.
- Monitor angina diary, headache.
- Do not crush MR tablets.

PATIENT EDUCATION

- Take in the morning; do not crush.
- Keep GTN spray for attacks.
- Avoid erectile dysfunction drugs.
- Rise slowly; limit alcohol.

KEY POINTS

- Long-acting nitrate; 100% bioavailable.
- Not for acute angina.
- MR once daily in the morning.
- No PDE-5 inhibitors.

SOURCES CHECKED **[359]** Isosorbide Mononitrate, oral (US label) **[360]** Isosorbide Mononitrate, oral (US label, Edenbridge Pharmaceuticals LLC) **[361]** isosorbide mononitrate, oral (US label)

Full references p. 432–457